

Standardization and Methodological Considerations for the Isometric Midthigh Pull

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ABSTRACT

THE ISOMETRIC MIDTHIGH PULL (IMTP) IS COMMONLY USED TO ASSESS AN ATHLETE'S FORCE GENERATION ABILITY. THIS TEST IS HIGHLY RELIABLE AND IS SIMPLE AND RELATIVELY QUICK TO PERFORM. THE DATA THAT CAN BE DETERMINED FROM THE FORCE-TIME CURVES GENERATED BY THE TEST HAVE BEEN SHOWN TO BE CLOSELY RELATED TO PERFORMANCE CAPACITIES IN A VARIETY OF DYNAMIC ATHLETIC TASKS. HOWEVER, WITHIN THE SCIENTIFIC LITERATURE, THERE ARE INCONSISTENCIES IN THE DATA COLLECTION PROCEDURES AND METHODS USED FOR DATA ANALYSIS THAT MAY IMPACT THE RESULTANT OUTPUT AND THE ABILITY TO COMPARE AND GENERALIZE RESULTS. THEREFORE, THE PRIMARY AIM OF THIS REVIEW IS TO IDENTIFY THE DIFFERENCES

IN IMTP TESTING PROCEDURES AND DATA ANALYSIS TECHNIQUES, WHEREAS IDENTIFYING THE POTENTIAL IMPACT THIS MAY HAVE ON THE DATA COLLECTED. THE SECONDARY AIM IS TO PROVIDE RECOMMENDATIONS FOR THE STANDARDIZATION OF TESTING PROCEDURES TO ENSURE THAT FUTURE IMTP DATA ARE OF MAXIMAL BENEFIT TO PRACTITIONERS AND RESEARCHERS. FOR A VIDEO ABSTRACT OF THIS ARTICLE, SEE SUPPLEMENTAL DIGITAL CONTENT 1 (SEE VIDEO, [HTTP://LINKS.LWW.COM/SCJ/A249](http://links.lww.com/scj/A249)).

INTRODUCTION

Maximal strength underpins performance in many athletic tasks (15,55,63) and as such, monitoring strength, usually by repetition maximum (RM) testing, is commonly performed by practitioners and researchers. Although RM testing is reliable (12,24,28), it can be perceived as fatiguing, posing an increased potential for injury risk, and only providing

information related to the maximal load lifted. By contrast, isometric testing, such as the isometric midthigh pull (IMTP), is potentially safer (18), less fatiguing, and allows for the quantification of peak force (PF), force at a variety of epochs, and can provide several measures of the rate of force development (RFD) (11,21,26,30,32,33). The diagnostic ability of these measures may be of importance when considering time-constrained tasks within sports, such as jumping, sprinting, and change of direction. Importantly, the IMTP has been shown to be highly reliable both within and between sessions, with low variability and low measurement error (8,11,18,24,26,27,32).

Performance in the IMTP has been associated with performance in numerous athletic tasks (7,18,30,33,40,41,45,46,49,59,64,66,67,69,72,73). Specifically, absolute PF has been associated with weightlifting

KEY WORDS:

force; rate of force development; posture; isometric strength

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Table 1
Relationships between peak force and performance in other activities

Author(s)	Subjects	1RM	Sprint	Jump	Change of direction	Other
Haff et al. (39)	8 trained (>2 y) men			SJ PF: $r = 0.76$		Force during dynamic MTP
	1RM PC = $1.21 \text{ kg} \cdot \text{kg}^{-1}$					90% 1RM: $r = 0.77$
						100% 1RM: $r = 0.80$
Stone et al. (60)	30 competitive sprint cyclists			CMJ height: $r = 0.59$		Absolute PF and sprint cycling performances: $r = 0.49\text{--}0.55$
				CMJ PP: $r = 0.79$		Relative PF and sprint cycling performances: $r = 0.45\text{--}0.60$
				SJ height: $r = 0.51$		AS PF and sprint cycling performances: $r = 0.45\text{--}0.58$
				SJ PP: $r = 0.78$		
Haff et al. (30)	6 elite women weightlifters	Snatch: $r = 0.93$		CMJ PP: $r = 0.88$		
				SJ PP: $r = 0.92$		
Kawamori et al. (39)	8 male collegiate weightlifters			CMJ PF: $r = 0.87$		Force during dynamic MTP
	1RM PC = $1.39 \text{ kg} \cdot \text{kg}^{-1}$			CMJ PRFD: $r = 0.85$		90% 1RM: $r = 0.82$
				CMJ PP: $r = 0.95$		
				CMJ height: $r = 0.82$		
				SJ height: $r = 0.87$		
McGuigan et al. (47)	8 Division III collegiate wrestlers	PC: $r = 0.97$ squat: $r = 0.96$ BP: $r = 0.73$				
McGuigan and Winchester (45)	22 college football players	PC, Squat, BP: $r = 0.61\text{--}0.72^a$				
	1RM PC = $1.11 \text{ kg} \cdot \text{kg}^{-1}$					

Table 1
(continued)

	1RM squat = 1.75 kg·kg ⁻¹					
Nuzzo et al. (49)	12 Division I collegiate athletes	PC: $r = 0.74$		CMJ PP: $r = 0.75$		
	1RM PC = 1.28 kg·kg ⁻¹			Relative PF and CMJ height: $r = 0.59$		
	1RM squat = 1.91 kg·kg ⁻¹					
Kraska et al. (41)	41 female and 22 male collegiate athletes			SJ: $r = 0.40$		
				SJ20: $r = 0.55$		
				CMJ: $r = 0.36$		
				CMJ20: $r = 0.55$		
				AS PF:		
				SJ: $r = 0.47$		
				SJ20: $r = 0.52$		
				CMJ: $r = 0.41$		
				CMJ20: $r = 0.52$		
Whittington et al. (72)	7 NCAA Division I track and field athletes					Ball throw distance
						PF: $r = 0.89$
						AS PF: $r = 0.91$
McGuigan et al. (46)	26 recreationally trained men	Squat: $r = 0.97$		CMJ height: $r = 0.72$		
	1RM squat = 1.30 kg·kg ⁻¹	BP: $r = 0.99$				
Khamoui et al. (40)	19 recreationally trained men			Relative PF and CMJ height: $r = 0.61$		Relative PF and high pull PV: $r = -0.60$
West et al. (71)	39 professional rugby league players		Relative PF and 10-m sprint time: $r = 0.37$	Relative PF and CMJ height: $r = 0.45$		
(continued)						

Table 1
(continued)

Spiteri et al. (59)	12 competitive female basketball players	IMTP relative PF, back squat: $r = 0.81$			T test: $r = -0.85$	
					505 COD = -0.79	
Winchester et al. (73)	26 recreationally trained men	Squat: $r = 0.97$		CMJ height: $r = 0.72$		
	1RM squat = $1.30 \text{ kg} \cdot \text{kg}^{-1}$	BP: $r = 0.99$				
Secomb et al. (53)	15 elite surfers			CMJ height: $r = 0.65$		
				SJ height: $r = 0.58$		
Beckham et al. (7)	12 collegiate national-level weightlifters	Snatch: $r = 0.83$				
		Clean and jerk: $r = 0.84$				
		Total: $r = 0.84$				
Thomas et al. (64)	14 collegiate team sport athletes		5 m: $r = -0.57$		505mod: $r = -0.57$	
			20 m: $r = -0.69$			
Thomas et al. (67)	22 collegiate team sport athletes			CMJ PF: $r = 0.45$		
Wang et al. (69)	15 collegiate rugby players	Squat: $r = 0.866$				

^aIndividual correlations not reported.

505mod = modified 505 change of direction; AS = allometrically scaled; BP = bench press; CMJ = countermovement jump; COD, change of direction; IMTP = isometric midhigh pull; PC = power clean; PF = peak force; PP = peak power; PRFD = peak rate of force development; PV = peak velocity; SJ = squat jump.

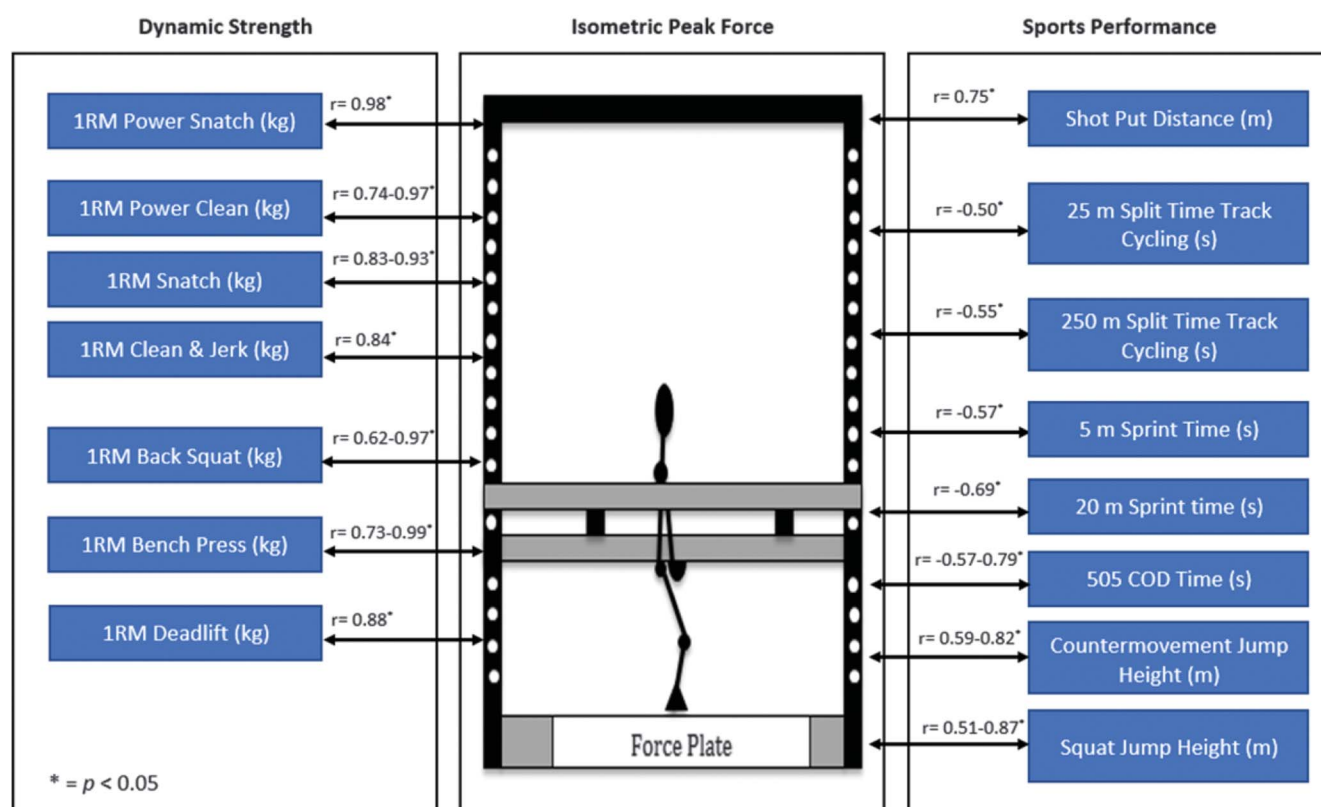


Figure 1. Relationships between isometric midhigh pull peak force and performance in other tasks. COD, change of direction.

performance (7,30), 1RM squat and power clean (45–47,49,59,69,73), 1RM deadlift (18), vertical jump performance (39–41,53,60,64,67), short sprint and change of direction times (59,64), sprint cycling performance (60), and throwing performance (72) (Table 1). By contrast, West et al. (71) reported no meaningful relationships between absolute PF and short sprint times or jump height, although they did observe large correlations between relative PF (PF/body weight) and these variables in rugby league players. Similarly, Nuzzo et al. (49) reported only a small relationship between absolute PF and jump height but a large relationship between relative PF and jump height (Table 1). The range of associations between PF and performance in other tasks is summarized in Figure 1. Researchers have also reported relationships between allometrically scaled PF and performance in athletic tasks (60,72), demonstrating similar correlations with those observed when ratio scaling is used (60).

Another way to examine the isometric force–time curve is to measure force at specific time epochs (e.g., 50–250 ms). It has been reported that these time-specific forces are associated with squat jump (SJ) and countermovement jump (CMJ) height (force at 50, 90, 250 ms) (41), weightlifting performance (force at 100, 150, 200, 250 ms) (7), and 1RM back squat (90–250 ms) (69). In addition, allometrically scaled force at 150 ms was reported to be related to mean and maximum club head speed during a golf swing (42), with allometrically scaled force at 50, 90, and 250 ms also related to jump performance (41) (Table 2). By contrast, however, force at 30–250 ms was not related to 1RM deadlift performance (18).

Equivocal results regarding the relationships between measures of RFD and performance in dynamic athletic tasks have been reported in the scientific literature. When examining how the RFD is quantified, 2 main methods exist within the literature (32). The first

method is to quantify the peak RFD (PRFD) that occurs during the IMTP with a predefined moving window, most typically lasting between 2 and 40 ms (32) (Table 3). When this method is used for analyzing the force–time curve, conflicting results exist within the scientific literature with some authors reporting significant relationships between the RFD and dynamic performance activities (30,33,39,41), whereas others report no meaningful relationship with 1RM performance (7,45–47), or SJ and CMJ performances (40,49,67). These differences may be attributable to the moving window, with Maffiuletti et al. (43) cautioning against the use of short windows (e.g., 2 ms), because they may be too sensitive to unsystematic variability and therefore less reliable. The second method for evaluating the RFD is to examine time-dependent epochs (32). The use of time-dependent epochs has been shown to be an effective method for examining the RFD

Table 2
Relationships between time-specific force and performance in other activities

Author(s)	Subjects	1RM	Sprint	Jump	Other
Kraska et al. (41)	41 female and 22 male collegiate athletes			PF50	
				SJ: $r = 0.33$	
				SJ20: $r = 0.52$	
				CMJ: $r = 0.27$	
				CMJ20: $r = 0.50$	
				AS PF50:	
				SJ: $r = 0.33$	
				SJ20: $r = 0.48$	
				CMJ20: $r = 0.45$	
				PF90	
				SJ20: $r = 0.37$	
				CMJ20: $r = 0.33$	
				AS PF90:	
				CMJ20: $r = 0.48$	
				PF250	
				SJ: $r = 0.39$	
				SJ20: $r = 0.56$	
				CMJ: $r = 0.34$	
				CMJ20: $r = 0.54$	
				AS PF250	
				SJ: $r = 0.42$	
				SJ20: $r = 0.51$	
				CMJ: $r = 0.34$	
				CMJ20: $r = 0.48$	
Beckham et al. (7)	12 collegiate national-level weightlifters	F100			
		Snatch: $r = 0.65$			
		Clean and jerk: $r = 0.64$			
		Combined total: $r = 0.65$			

Table 2
(continued)

		F150			
		Snatch: $r = 0.64$			
		Clean and jerk: $r = 0.61$			
		Combined total: $r = 0.62$			
		F200			
		Snatch: $r = 0.73$			
		Clean and jerk: $r = 0.71$			
		Combined total: $r = 0.72$			
		F250			
		Snatch: $r = 0.80$			
		Clean and jerk: $r = 0.80$			
		Combined total: $r = 0.80$			
West et al. (71)	39 professional rugby league players		F100 and 10 m: $r = -0.66$	F100 and CMJ PP: $r = 0.55$	
			Relative F100 and 10 m: $r = -0.68$	Relative F100 and CMJ PP: $r = 0.38$	
				Relative F100 and CMJ height: $r = 0.43$	
Wang et al. (69)	15 collegiate rugby players	Squat			
		F90: $r = 0.76$			
		F100: $r = 0.78$			
		F150: $r = 0.78$			
		F200: $r = 0.77$			
		F250: $r = 0.82$			
Leary et al. (42)	12 recreational golfers				Golf Club Head Speed
					ASF150 and mean club head speed: $r = 0.46$
					ASF150 and Max club head speed: $r = 0.47$

AS = allometrically scaled; CMJ20 = countermovement jump with 20 kg; F100 = force at 100 ms; F150 = force at 150 ms; F200 = force at 200 ms; F250 = force at 250 ms; F90 = force at 90 ms; SJ20 = squat jump with 20 kg.

during the IMTP and relating it to various sports performance tasks. For example, Spiteri et al. (58) report that

athletes who produce higher RFD to 90 and 100 ms are able to demonstrate faster agility times during a 45° cutting

task. One possible explanation why some RFD measures relate to dynamic performance activities and others do

Table 3
Relationships between RFD and performance in other activities

Author(s)	Subjects	1RM	Sprint	Jump	Change of direction	Other
Haff et al. (33)	8 trained (>2 y) men			PRFD		RFD during dynamic MTP
	1RM PC = 1.21 kg · kg ⁻¹			SJ power: <i>r</i> = 0.76		80% 1RM: <i>r</i> = 0.84
				SJ height: <i>r</i> = 0.82		90% 1RM: <i>r</i> = 0.88
						100% 1RM: <i>r</i> = 0.84
Haff et al. (30)	6 elite women weightlifters	PRFD		PRFD		
		Snatch: <i>r</i> = 0.79		CMJ PP: <i>r</i> = 0.81		
		Combined Total: <i>r</i> = 0.80		SJ PP: <i>r</i> = 0.84		
McGuigan et al. (47)	8 Division III collegiate wrestlers					PRFD and coaching ranking: <i>r</i> = 0.62
Kawamori et al. (39)	8 male collegiate weightlifters					Force during dynamic MTP
	1RM PC = 1.39 kg · kg ⁻¹					90% 1RM: <i>r</i> = 0.69
						120% 1RM: <i>r</i> = 0.74
Nuzzo et al. (49)	12 Division I collegiate athletes			PRFD		
	1RM PC = 1.28 kg · kg ⁻¹			CMJ PP: <i>r</i> = 0.65		
	1RM squat = 1.91 kg · kg ⁻¹					
Kraska et al. (41)	41 female and 22 male collegiate athletes			PRFD		
				SJ: <i>r</i> = 0.48		
				SJ20: <i>r</i> = 0.66		
				CMJ: <i>r</i> = 0.43		
				CMJ20: <i>r</i> = 0.62		
Whittington et al. (72)	7 NCAA Division I track and field athletes					Ball throw distance: <i>r</i> = 0.78
Khamoui et al. (40)	19 recreationally trained men					RFD50 and high pull PV: <i>r</i> = 0.56
						RFD100 and high pull PV: <i>r</i> = 0.56

Table 3
(continued)

West et al. (71)	39 professional rugby league players		PRFD	PRFD		
			10 m: $r = -0.66$	CMJ height: $r = 0.39$		
Beckham et al. (7)	12 collegiate national-level weightlifters	RFD200				
		Snatch: $r = 0.65$				
		Combined Total: $r = 0.60$				
		RFD250				
		Snatch: $r = 0.78$				
		Clean and jerk: $r = 0.72$				
		Combined Total: $r = 0.75$				
Thomas et al. (64)	14 collegiate team sport athletes		PRFD		PRFD	
			5 m: $r = -0.58$		505mod: $r = -0.57$	
			20 m: $r = 0.71$			
Wang et al. (69)	15 collegiate rugby players		5 m:		Proagility:	
			PRFD: $r = -0.54$		PRFD: $r = -0.52$	
			RFD30: $r = 0.57$		RFD30: $r = 0.52$	
			RFD50: $r = 0.53$		RFD50: $r = 0.53$	
					RFD90: $r = 0.53$	
					RFD100: $r = 0.52$	

PRFD = peak RFD; PV = peak velocity; RFD100 = mean RFD between 0 and 100 ms; RFD200 = mean RFD between 0 and 200 ms; RFD250 = mean RFD between 0 and 250 ms; RFD30 = mean RFD between 0 and 30 ms; RFD50 = mean RFD between 0 and 50 ms; RFD90 = mean RFD between 0 and 90 ms.

not is the method of calculation and reliability of the method. For example, Haff et al. (32) have shown that the only PRFD measure that is reliable is when a 20-ms moving window is used, supporting previous suggestions by Maffiuletti et al. (43). Conversely, using time-dependent epochs such as 0–90, 0–150, 0–200, and 0–250 ms to calculate the mean RFD across the specific duration produces much more reliable results and generally have better relationships with dynamic performance measures. Therefore, it is generally recommended that using time-specific RFD epochs is warranted when using the IMTP as a performance diagnostic tool (32).

Another method for analyzing the force-time curve derived from an IMTP is to examine the isometric impulse (67,68). For example, impulse values across different epochs (0–100, 0–200, and 0–300 ms) have been associated with 5- and 20-m sprint times as well as 505 change of direction times (64), PF and power during the SJ and CMJ (68) (Table 4). Although determining the isometric impulse of various epochs within the force-time curve achieved during the IMTP yields useful information, much more research is needed to understand how best to use this measurement in a sports performance monitoring program.

The PF achieved during the IMTP has also been used to monitor adaptations to training (5,36,50,51,57,70,74), with some authors also including RFD (36,51,52,74). Peak force and PRFD have also been used in an attempt to identify levels of fatigue or recovery (4,29,35,44). More recently, researchers have started to investigate the potential of the IMTP to investigate between-limb asymmetries, using dual force platforms (1–3) and a unilateral stance IMTP (25,65). In addition, the PF during the IMTP has been divided by the PF during an SJ or CMJ, to calculate the dynamic strength index (ratio of PF during the CMJ or SJ and IMTP PF), in attempt to identify whether an athlete needs to focus more on maximal force production or rapid dynamic force production (14,52,54,56,66).

VARIATION IN TESTING AND DATA ANALYSIS PROCEDURES

Unfortunately, there is substantial variation across testing protocols reported within the scientific literature, including differences in knee and hip joint angles (120–150° and 124–175°, respectively), sampling frequency (500–2000 Hz), pull onset identification thresholds including absolute (20–75 N) and relative (2.5–10% body weight) threshold values, and smoothing and filtering approaches, with some authors not stating hip angles, thresholds, or filtering procedures (Table 5). In addition, if practitioners or researchers are intending to use published values for comparison, they should be mindful that some data are presented as net force (gross force – body weight), whereas others report gross measures, along with ratio and allometric scaling used in some studies. These 2 latter approaches may impact the results less, as allometric scaling uses an exponent related to body mass, (13) although allometric scaling will reduce the resultant values compared with ratio scaling, with greater variation introduced depending on the exponent used (Table 5).

Numerous authors have suggested that the posture adopted during the IMTP should replicate the start of the second pull phase of the clean, (30,31,33,60); however, only 2 studies have actually assessed the participants knee joint angles during the clean and then adopted these angles during the IMTP (30,31). This is most likely due to time and practicality of assessing specific joint angles during the clean before performing the IMTP, especially when assessing large squads of athletes. Interestingly, hip joint angles were not reported within these 2 studies (30,31).

Because of the variety of knee and hip joint angles reported within the literature, Comfort et al. (11) investigated a range of knee (120, 130, 140, and 150°) and hip (125 and 145°) joint angles, along with self-selected posture (knee $133 \pm 3^\circ$ and hip $138 \pm 4^\circ$) based on the athletes' preferred position to start the second pull of a clean, which is what the posture

adopted during the IMTP was originally based on (33). The results of the study indicated that there were no significant or meaningful differences in PF, PRFD, or impulse between postures, although the preferred (self-selected) posture demonstrated the highest reliability and the lowest measurement error. By contrast, Beckham et al. (6) found that powerlifters produced greater PF during an isometric testing with a vertical torso compared with a deadlift-specific body position at the same bar height, described as being a “relatively straight-legged position and somewhat bent over the bar”. The authors suggested that the upright position may have provided a mechanical advantage and a posture more optimal for force production against the bar. In another study, Beckham et al. (8) compared the effects of different hip joint angles (125 versus 145°), while standardizing the knee joint angle (125°) reporting meaningful and significantly higher PF and force at different epochs (50, 90, 200, and 250 ms) in the more upright (145°) position, especially in subjects with greater experience in performing weightlifting exercises and their derivatives, in contrast to Comfort et al. (11). Interestingly, Beckham et al. (8) reported small changes in joint angles throughout the execution of the test and based on these observations recommend that, in the future, researchers and practitioners should adopt standardized knee and hip angles of 120–135° and 140–150°, respectively.

More recently, Dos'Santos et al. (26) compared hip joint angles of 145 and 175° with a standardized knee joint angle of 145°, finding greater time-specific force values and RFD at predetermined epochs, with a 145° hip angle (Table 5). The hip angle of 175° previously reported by Kraska et al. (41) and replicated by Beckham et al. (6) actually refer to trunk angle relative to vertical, to ensure an upright trunk (forward lean of 5° from vertical), exhibiting an upright trunk as previously described (30,31,33,60) rather than a 175° hip angle as used by Dos'Santos et al. (26). The authors of a recent meta-analysis also highlight the fact that practitioners should carefully

Table 4
Relationships between time-specific impulse and performance in other activities

Author(s)	Subjects	Sprint	Jump	Change of direction
Thomas et al. (64)	14 collegiate team sport athletes	Imp100		Imp100, 505mod: $r = -0.58$
		5 m: $r = -0.71$		Imp300, 505mod: $r = -0.62$
		20 m: $r = 0.75$		
		Imp300		
		5 m: $r = -0.74$		
		20 m: $r = 0.78$		
Thomas et al. (67)	22 collegiate team sport athletes		Imp100	
			SJ PF: $r = 0.57$	
			SJ PP: $r = 0.60$	
			CMJ PF: $r = 0.64$	
			CMJ PP: $r = 0.51$	
			Imp200	
			SJ PF: $r = 0.56$	
			SJ PP: $r = 0.59$	
			CMJ PF: $r = 0.63$	
			CMJ PP: $r = 0.50$	
			Imp300	
			SJ PF: $r = 0.58$	
			SJ PP: $r = 0.60$	
			CMJ PF: $r = 0.63$	
			CMJ PP: $r = 0.49$	

CMJ = countermovement jump; Imp100 = impulse over 100 ms; Imp200 = impulse over 200 ms; Imp300 = impulse over 300 ms; PF = peak force; PP = peak power; SJ = squat jump.

consider the specific protocol, including joint angles, to ensure repeatability of the measures (27).

While adopting standardized knee and hip angles during the IMTP may seem logical, this practice may place athletes in a suboptimal pulling position, because of the range of angles reported across individuals for the second pull phase of the clean (30,31). Therefore, it is best to consider the individual athlete's appropriate second pull position and then quantify the knee and hip angles. This practice allows for the individual athlete's anthropometrics to be considered and allows them to

assume an optimal pulling position, in line with the range of joint angles recommended by Beckham et al. (8). Once the pulling position is established, then it is recommended that practitioners and researchers ensure that the individual starting postures are replicated between trials and testing sessions. Joint angles should be assessed before the commencement of the pull because of slight changes in joint angles during the pull (68).

Haff et al. (32) suggest using minimal pretension before initiation of the pull, as this is likely to impact both time-specified force and RFD, with

Dos'Santos et al. (26) recently reporting that the 175° hip angle results in significantly higher "body weight" because of increased pretension, compared with a 145° hip angle, which may have contributed to the differences in time-specific force values and RFD that were reported. Similarly, Maffiuletti et al. (43) suggested that pretension is undesirable when assessing isometric RFD, albeit with a focus on single joint assessment; it would, therefore, be advantageous to visually inspect the force-time data before and after the isometric pull, to ensure that there are no differences in force, which should represent body weight.

Table 5
Reported testing and data analysis procedures

Author(s)	Knee angle	Hip angle	Sampling frequency	Onset threshold	Scaling	Smoothing and filtering	RFD calculation
Haff et al. (33)	144 ± 5°	145 ± 3°	500 Hz	—	Net force	—	PRFD (2 ms window)
Stone et al. (60)	140–145°	—	600 Hz	—	Net absolute, relative, and AS	—	PRFD (1.7 ms window)
Haff et al. (30)	127–145 ^{ob}	—	600 Hz	—	Net force	—	PRFD (1.7 ms window)
McGuigan et al. (47)	130°	—	500 Hz	—	Absolute	—	PRFD (2 ms window)
Kawamori et al. (39)	141 ± 10°	124 ± 11°	500 Hz	—	—	—	PRFD (2 ms window)
Haff et al. (31)	127–145 ^{ob}	—	600 Hz	—	Net force	—	PRFD (1.7 ms window)
Nuzzo et al. (49)	140°	—	1,000 Hz	—	Ratio	—	Mean RFD
Winchester et al. (74)	130°	—	—	—	Net	—	—
Winchester ^c et al. (73)	—	—	—	—	—	—	—
McGuigan and Winchester (45)	130°	—	960 Hz	—	—	—	—
							Assumed peak due to the values
Kraska et al. (41)	120–135°	170–175°	1,000 Hz	—	Absolute and AS	—	—
		In line with Haff et al. (1997) ^a					Assumed peak due to the values
Whittington et al. (72)	120–135° “Self-selected”	170–175° “self-selected”	1,000 Hz	—	—	—	PRFD (1 ms window)

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Table 5
(continued)

McGuigan et al. (46)	130°	—	960 Hz	—	—	—	—
					Assumed net due to the values		Assumed mean due to the values
West et al. (71)	120–130°	—	1,000 Hz	5 SD of mean force after trigger	Net	Dual-pass Butterworth filter (low pass, 20 Hz cut-off)	PRFD (1 ms window)
	In line with Haff et al. (2005), Stone et al. (2004) ^a						
Crewther et al. (16)	120–130°	—	1,000 Hz	—	Net	Dual-pass Butterworth filter (low pass, 20 Hz cut-off)	PRFD (1 ms window)
	In line with Haff et al. (2005), Stone et al. (2004) ^a						
Beckham et al. (6)	In line with Haff et al. (1997) and Kraska et al. (2009) ^a	In line with Haff et al. (1997) and Kraska et al. (2009) ^a	1,000 Hz	—	Absolute and AS	4th order Butterworth low-pass filter 100 Hz	Not included
Beckham et al. (7)	120–135°	175°	1,000 Hz	—	Absolute, ratio and AS	4th order Butterworth low-pass filter 100 Hz	Mean and PRFD (1 ms window)
Sheppard et al. (56)	130°	155–165°	600 Hz	—	Net	—	Not included
Comfort et al. (11)	120, 130, 140, 150° and self-selected (133 ± 3°)	125, 145° and self-selected (138 ± 4°)	600 Hz	40 N	Absolute	—	PRFD (1.7 ms window)
Thomas et al. (64)	Self-selected	Self-selected	600 Hz	—	Absolute	4th order Butterworth low-pass filter 16 Hz	PRFD (1.7 ms window)

(continued)

Table 5
(continued)

Thomas et al. (67)	Self-selected	Self-selected	600 Hz	—	Absolute and relative	4th order Butterworth low-pass filter 16 Hz	PRFD (1.7 ms window)
Thomas et al. (66)	Self-selected	Self-selected	600 Hz	—	Absolute	4th order Butterworth low-pass filter 16 Hz	Not included
Haff et al. (32)	140.0 ± 6.6°	137.6 ± 12.9°	1,000 Hz	—	Net	Rectangular smoothing with a moving half width of 12	PRFD (20 ms window)
							RFD _{30, 50, 90, 100, 150, 200, 250}
Secomb, et al. (52)	125–140°	—	600 Hz	—	Absolute and relative	—	Not included
Secomb et al. (53)	125–140°	—	600 Hz	—	Absolute and relative	—	Not included
Secomb et al. (54)	—	—	600 Hz	—	Absolute and relative	—	Not included
	Stated similar to Haff et al. (2005)	Stated similar to Haff et al. (2005)					
Tran et al. (68)	—	—	600 Hz	—	Absolute and relative (assumed net due to the values)	4th order Butterworth low-pass filter 10 Hz	
	Stated similar to Haff et al. (1997)	Stated similar to Haff et al. (1997)					
Spiteri et al. (58)	140°	140°	2,000 Hz	—	Relative	—	RFD _{30, 50, 90, 100}
Sjokvist et al. (57)	states in line with Stone et al. (2004)				Absolute and relative	—	Not included
Welch et al. (70)	No specific detail provided other than bar positioned at midhigh				Relative	—	Not included
Wang et al. (69)	Self-selected	Self-selected	1,000 Hz	—	Net	—	PRFD (20 ms window)

Table 5
(continued)

							RFD _{30r} 50, 90, 100, 150, 200, 250
Mangine et al. (44)	Self-selected	Self-selected	1,000 Hz	—	Net	—	PRFD (20 ms window)
							RFD _{30r} 50, 90, 100, 150, 200, 250
Halperin et al. (34)	130–140°	Not stated	1,000 Hz	—	—	—	Not included
Dos’Santos et al. (22)	Self-selected	Self-selected	2,000 Hz (down- sampled to 1,500, 1,000, and 500 Hz)	75 N	Absolute	20-ms moving average	RFD ₁₀₀
							RFD ₁₅₀
							RFD ₂₀₀
Bartolomei et al. (4)	140°	125°	1,000 Hz	—	Absolute	—	PRFD (20 ms window)
James et al. (38)	141.9 ± 4.3°	139.2 ± 4.1°	1,000 Hz down- sampled to 100 Hz to compare with strain gauge	20 N	Net	4 th order Butterworth low-pass filter 10 Hz	PRFD (20 ms window)
							RFD _{30r} 50, 90, 100, 150, 200, 250
De Witt et al. (18)	144 ± 3°	137 ± 3°	1,000 Hz	—	—	—	PRFD (20 ms window)
					Assumed net due to the values		RFD _{30r} 50, 90, 100, 150, 200, 250
Dos’Santos, Thomas et al. (24)	137–146 ^{od}	140–149 ^{od}	1,000 Hz	40 N	Absolute	—	Not included
Dos’Santos, et al. (21)	Self-selected	Self-selected	1,000 Hz	2.5% BW, 5% BW, 10% BW, >75 N, 5 SD BW	Absolute	—	RFD ₁₀₀
							RFD ₁₅₀
							RFD ₂₀₀

(continued)

Table 5
(continued)

Beckham et al. (8)	125°	125 and 145°	1,000 Hz	—	Absolute and AS	2nd order Butterworth low-pass filter 10 Hz	Not included
Oranchuk et al. (50)	135–145°	—	1,000 Hz	2.5% of mean body mass, based on force–time data	Relative	4th order Butterworth filter, with 20-Hz cut-off	PRFD (20 ms window)
Dobbin et al. (20)	140°	Self-selected, shoulder above the bar (as described by Thomas et al. 2015)	1,200 Hz	—	Net relative and AS	—	Not included
Beattie et al. (5)	131 ± 9°	—	1,000 Hz	—	Relative	—	Not included
Dos'Santos et al. (26)	145°	145° and 175°	1,000 Hz	5 SD BW	Net	Unfiltered	PRFD
							RFD ₁₀₀
							RFD ₁₅₀
							RFD ₂₀₀
Leary et al. (42)	142 ± 7°	146 ± 11°	1,000 Hz	—		Rectangular smoothing with a moving half width of 12	PRFD
							RFD _{30, 50, 90, 100, 150, 200, 250}

— = not stated.

Net force = gross force – body weight.

Mean force (change in force/change in time from onset of force production to time to peak force).

^aIncorrectly cites joint angles “in line with previous research” when the referenced studies used different joint angles.^bBased on knee angle achieved during the second pull phase of the clean for each individual.^cPublished abstract.^dSelf-selected to replicate the start of the second pull.BW = body weight (during the initial period of quiet standing), PRFD = Peak instantaneous RFD (the greatest rate of change in force between 2 tangential points; RFD₁₀₀ = subscript numbers refer to the epoch for mean RFD; the window differs based on sampling frequency).

Interestingly, numerous authors state that they have adopted the postures previously reported by other researchers but, in fact, report different angles to those stated in the studies that they cite, or cite multiple researchers who reported different postures (Table 5). These differing postures are most likely related to individual athlete anthropometric profiles. It is therefore important that researchers carefully report and justify their choice of joint angles but, more importantly, standardize these between trials and testing sessions.

Other researchers have used strain gauge-based equipment, with the handle attached using a chain (16,17,37,38,48) with a range of sampling frequencies (100–133 Hz (17,37,38)) and joint angles (knee 120–130° (17), $142 \pm 4^\circ$ (38), $143 \pm 7^\circ$ (37), 160° (48); hip $139 \pm 4^\circ$ (38), $144 \pm 5^\circ$ (37)). However, findings of 2 research groups that compared strain gauge systems with a force platform demonstrated that the strain gauge significantly underestimated PF, by ~8% (38) to ~10% (20). In addition, James et al. (38) found that measures of RFD did not meet acceptable standards of reliability. Although such systems can measure PF, which can be ratio or allometrically scaled, there does not seem to be an effective way to accurately measure or calculate RFD and are therefore not recommended if practitioners have access to a force platform.

RECOMMENDATIONS FOR CORRECT ISOMETRIC MIDTHIGH PULL ASSESSMENT

Because of the noticeable variations in assessment procedures, including posture, sampling frequency, and methods of calculating specific variables (namely use of different sampling frequencies, onset thresholds, and the method for the calculation of RFD), we suggest appropriate standardization of all testing procedures for the IMTP. Such standardization should permit more meaningful comparisons of individual performances between testing sessions, comparisons between athletes, and more effective comparisons between published studies. Standardization should also include the

verbal cues because attentional focus has been shown to affect force production, with an external focus of “push as hard and fast as possible” resulting in greater PF compared with an internal focus (34).

RECOMMENDED TESTING PROCEDURES

Before initiation of IMTP testing, the bar height necessary to obtain the correct body position should be determined. This should be an iterative process in which the athlete starts with a bar height that allows the athlete to assume a body position that replicates the start of the second pull position during the clean. The bar height should then be adjusted up or down to allow the athlete to obtain the optimal knee (125–145°) and hip (140–150°) angles (6,8,26). The body position should be very similar to the second pull of the clean and the clean grip midthigh pull exercise (19): upright torso, slight flexion in the knee resulting in some dorsiflexion, shoulder girdle retracted and depressed, shoulders above or slightly behind the vertical plane of the bar, feet roughly centered under the bar approximately hip width apart, knees underneath and in front of the bar, and thighs in contact with the bar (close to the inguinal crease dependent on limb lengths) (Figure 2). When making joint measurements, the athlete should ensure that no tension is applied to the bar, but

that all “slack” (e.g., elbow flexion and shoulder girdle elevation/protraction) is removed from the body because this would result in a change in joint angles during the maximal effort that is undesirable (8).

Although the use of a “self-selected” body position is likely beneficial to efficiency of testing, it is not recommended without ensuring that the hip and knee joint angles fall within the ranges recommended above, because of the influence of body positioning on force generation (6,8,26). The bar height used and joint angles obtained should be recorded, so that repeated measurements can be standardized and therefore replicate the individuals’ body position between sessions, ensuring that differing results in subsequent testing are not the result of changed body position (8,26). It is also considered best practice to measure the individual’s grip width and foot position and standardize these for individuals across sessions (unless working with youth athletes where changes in stature as a result of maturation may require increased stance and grip width) as each can affect body positioning relative to the bar (19). After the bar height and posture have been established, a short familiarization session of submaximal trials is recommended approximately 48 hours before testing (e.g., 3×3 -second trials,



Figure 2. Correct posture for the isometric midthigh pull, illustrating an upright trunk, replicating the start position of the second pull of the clean.

Standardization of the Isometric Midthigh Pull

each of 50, 75, and 90% of perceived maximum effort). Although a consensus on the optimal amount of familiarization has not yet been reached, nearly all IMTP studies use some familiarization.

Athletes should complete some manner of standard generalized warm-up (62). Although there is variability in the generalized warm-up chosen among studies, most studies use a warm-up that incorporates clean derivatives, such as the dynamic midthigh pull, and should thus be a component of the standard warm-up (7,21,24,32,33). Submaximal trials of the IMTP are also recommended before maximal effort trials (e.g., 3 seconds each of 50% maximal effort, 75% maximal effort, and 90% maximal effort, separated by 60-second rest). During this time, the athlete should be secured to the bar using lifting straps and athletic tape to ensure that grip strength is not a limiting factor (Figure 3) (30,33).

For each of the maximal effort trials, standardized instructions should be given to the athlete of some iteration of “push your feet into the ground as fast and as hard as possible” to ensure that both maximal RFD and PF are obtained (10,34). It is essential that athletes understand that the focus is to drive the feet directly into the force platform and not attempt to pull the bar with the arms, or rise up on to their toes. The athlete should get into the correct body position for the IMTP, using just enough pretension to achieve the correct body position and remove “slack” from the body, but without any more pretension than is necessary to get the “quiet standing” necessary for a stable force baseline (43). This can be verified by monitoring the athlete’s body positioning and ensuring that the force trace created by the athlete is both similar to body mass and steady, with trials where a change in force >50 N occurs during this period rejected (21). This should be explained to the athletes, and they should be encouraged to stay as still as possible during this period to accurately determine body weight and onset threshold.

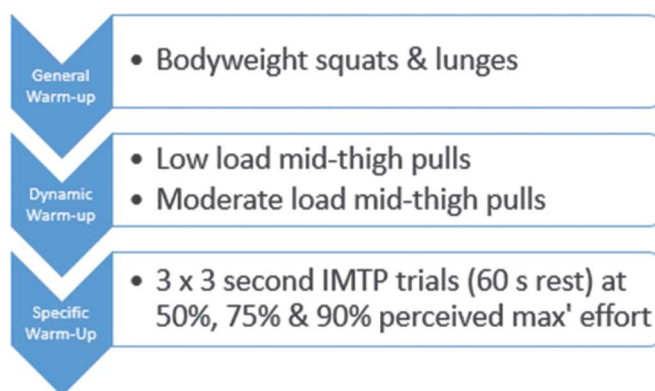


Figure 3. Standardized warm-up procedure.

A countdown of “3, 2, 1, PULL!” gives the athlete sufficient warning to be ready to give a maximum effort and provides at least one second of quiet standing to enable the identification of the onset of the pull (Figure 5A). Strong verbal encouragement from researchers and teammates ensures that the athlete gives a maximum effort (9). A minimum of 2 trials should be collected, provided that each of those trials have no errors by the athlete (e.g., countermovement, excessive pretension, and leaning on the bar before the pull [Figure 4]). With increasing PF, additional trials should be performed, until the PF values of the trials are separated by <250 N (30,33). It is noted, however, that a percentage of

PF may be advantageous because an absolute value will affect stronger and weaker athletes differently, although the exact effect of this has not been investigated.

Visual inspection of the force–time curves during testing can easily be used to determine whether the trials are acceptable, or whether additional trials should be performed. In addition to the trials being within 250 N between attempts, trials should be repeated if there is not a stable weighing period (clear fluctuation in the force–time data) or a clear countermovement before the initiation of the pull (Figure 5C) because this will interfere with accurate identification of the

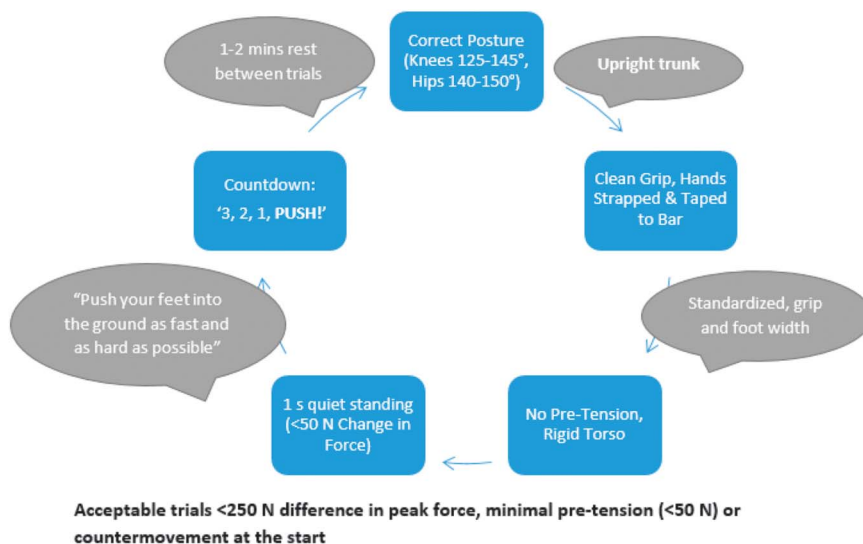


Figure 4. Standardized isometric midthigh pull testing procedure.

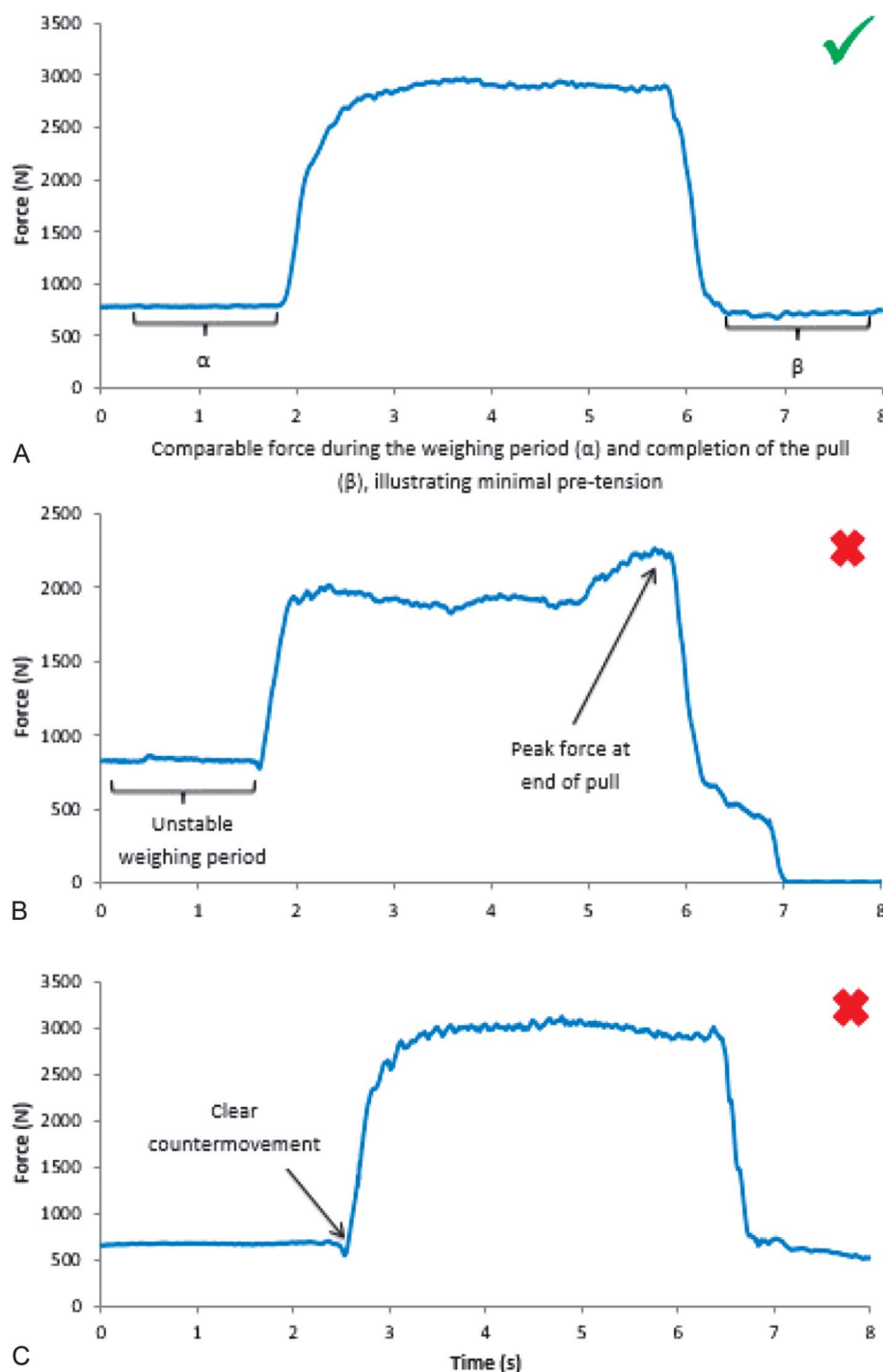


Figure 5. Examples of acceptable and unacceptable isometric midhigh pull force–time traces. A) Acceptable, B & C) Unacceptable force–time traces.

initiation of the pull (19), or if the PF occurs at the end of the trial (Figure 5B). It is also important to check that the force during the initial period of quiet standing (in the ready position, strapped to the bar, and

immediately before commencing the pull) represents body weight, and therefore, no previous tension has been applied (Figure 5A) because this will interfere with pull onset identification (19).

RECOMMENDED DATA ANALYSIS AND REPORTING

Collection of IMTP force–time data can be compiled accurately with a sampling frequency as low as 500 Hz, but if higher sampling frequencies can be used, then

they are preferred as they may increase the accuracy of time-dependent measures (21). Specifically, the utilization of frequencies $\geq 1,000$ Hz are recommended especially if early force-time variables are of interest (e.g., force at 50 or 100 ms) (21). There are not enough data for a consensus regarding optimal filtering and/or smoothing methods for the IMTP (23), although unfiltered data have been suggested as optimal for analysis of CMJ performance (61) and where possible, unfiltered data for isometric testing (23,43). It is therefore suggested that unfiltered and non-smoothed data are used for subsequent analysis (23) because most of the RFD and impulse characteristics are dependent on an accurate determination of the start of the pull (21), although data from portable force platforms may exhibit greater “noise” and warrant smoothing. Accurate identification of the start of the inflection point is often achieved using automated methods—we recommend using 5 SDs of body weight during an initial 1-second weighing period before the (usually 1 second) quiet standing (in the ready position, strapped to the bar, immediately before commencing the pull) as the threshold for determining the onset of the pull (21), although this may vary with technical idiosyncrasies of different force platforms (e.g., noise magnitude). Trials that do not have a stable baseline force trace during the weighing period (change in force >50 N) should be rejected and subsequently another trial should be performed (21,43) (Figure 5). To facilitate this stable period, it is essential to enforce and practice this during the warm-up/familiarization trials.

It is recommended that time-specific RFD epochs (50, 100, 150, 200, and 250 ms commonly reported) should be used when using the IMTP as a sport performance diagnostic tool as these are not only reliable (32) but can be selected specific to the durations relevant to the specific sporting tasks, such as ground contact time during acceleration or peak running speeds. By contrast,

maximal strength capabilities can be inferred from PF (Table 1).

When reporting results from IMTP testing, it is important that the hip and knee angles used by each athlete, to establish the bar height, be reported (8,26). Such standardization of posture between trials and testing sessions ensures that data are comparable between sessions, groups of athletes, and studies (8,26). Although there is no consensus as to the superiority of either net or gross force values for the IMTP, it is important that researchers report whether body weight was or was not included in the force and impulse values reported (7). Other methodological considerations, such as the method for identifying the onset of the pull (and threshold) (21), methods used for smoothing/filtering force platform data (23), sampling frequency, and other aspects of analysis (22), such as the exponent used for allometric scaling, should be reported because each are important for accurately interpreting results from the study.

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